

DHCP CONFIGURATION

Assign Ip address in all cisco routers

R1					R2				
do sh ip int br					do sh ip int br				
Interface	IP-Address	OK?	Method	Status	Interface	IP-Address	OK?	Method	Status
Ethernet0/0	50.0.0.1	YES	manual	up	Ethernet0/0	192.168.2.1	YES	manual	up
Ethernet0/1	172.16.0.1	YES	manual	up	Ethernet0/1	172.16.0.2	YES	manual	up
Ethernet0/2	unassigned	YES	TFTP	down	Ethernet0/2	unassigned	YES	TFTP	down
Ethernet0/3	unassigned	YES	TFTP	down	Ethernet0/3	unassigned	YES	TFTP	down
R1(config)#					R2(config)#				

Add static route in R1 & R2

R1					R2				
Ethernet0/3 unassigned YES TFTP down					Ethernet0/3 unassigned YES TFTP down				
R1(config)#do sh ip route					R2(config)#do sh ip route				
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP					Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP				
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area					D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area				
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2					N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2				
E1 - OSPF external type 1, E2 - OSPF external type 2					E1 - OSPF external type 1, E2 - OSPF external type 2				
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2					i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2				
ia - IS-IS inter area, * - candidate default, U - per-user static route					ia - IS-IS inter area, * - candidate default, U - per-user static route				
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP					o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP				
a - application route					a - application route				
+ - replicated route, % - next hop override, p - overrides from PFR					+ - replicated route, % - next hop override, p - overrides from PFR				
Gateway of last resort is not set					Gateway of last resort is not set				
50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks					S 50.0.0.0/8 [1/0] via 172.16.0.1				
C 50.0.0.0/8 is directly connected, Ethernet0/0					172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks				
C 50.0.0.1/32 is directly connected, Ethernet0/0					C 172.16.0.0/16 is directly connected, Ethernet0/1				
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks					L 172.16.0.2/32 is directly connected, Ethernet0/1				
C 172.16.0.0/16 is directly connected, Ethernet0/1					192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks				
L 172.16.0.1/32 is directly connected, Ethernet0/1					C 192.168.2.0/24 is directly connected, Ethernet0/0				
S 192.168.2.0/24 [1/0] via 172.16.0.2					L 192.168.2.1/32 is directly connected, Ethernet0/0				
R1(config)#					R2(config)#				

Enable dhcp server for HR department and Sales department in R1

R1				
R1(config)#				
R1(config)#ip dhcp pool HR				
R1(dhcp-config)#network 50.0.0.0 255.0.0.0				
R1(dhcp-config)#default-router 50.0.0.1				
R1(dhcp-config)#dns-server 8.8.8.8				
R1(dhcp-config)#lease 1 2 3				
R1(dhcp-config)#exit				
R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50				
R1(config)#exit				
R1#				
*Sep 2 08:26:48.157: %SYS-5-CONFIG_I: Configured from console by console				
R1#conf t				
Enter configuration commands, one per line. End with CNTL/Z.				
R1(config)#ip dhcp pool SALES				
R1(dhcp-config)#network 192.168.2.0 255.255.255.0				
R1(dhcp-config)#default-router 192.168.2.1				
R1(dhcp-config)#dns-server 8.8.8.8				
R1(dhcp-config)#lease 1 2 3				
R1(dhcp-config)#exit				
R1(config)#ip dhcp ex				
R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.50				
R1(config)#				

In vpc1 and vpc2 send a request to dhcp to get a ip

VPC-2	VPC-1
Modified version for EVE-NG. Press '?' to get help. Executing the startup file	Modified version for EVE-NG. Press '?' to get help. Executing the startup file
VPC-2> ip dhcp DDORA IP 50.0.0.52/8 GW 50.0.0.1	VPC-1> ip dhcp DDORA IP 50.0.0.51/8 GW 50.0.0.1
VPC-2> show ip	VPC-1> show ip
NAME : VPC-2[1] IP/MASK : 50.0.0.52/8 GATEWAY : 50.0.0.1 DNS : 8.8.8.8 DHCP SERVER : 50.0.0.1 DHCP LEASE : 93775, 93780/46890/82054 MAC : 00:50:79:66:68:2b LPORT : 20000 RHOST:PORT : 127.0.0.1:30000 MTU : 1500	NAME : VPC-1[1] IP/MASK : 50.0.0.51/8 GATEWAY : 50.0.0.1 DNS : 8.8.8.8 DHCP SERVER : 50.0.0.1 DHCP LEASE : 93765, 93780/46890/82054 MAC : 00:50:79:66:68:27 LPORT : 20000 RHOST:PORT : 127.0.0.1:30000 MTU : 1500
VPC-2> █	VPC-1> █

Enable dhcp relay agent on router 2 client connected interface

```
R2
R2(config)#intr
R2(config)#int eth0/0
R2(config-if)#ip help
R2(config-if)#ip helper-address 172.16.0.1
R2(config-if)#ex
R2(config)#
R2(config)# █
```

In vpc 3 and Win send a requests to dhcp to get a ip

VPC-4	QEMU (Win)
Modified version for EVE-NG. Press '?' to get help. Executing the startup file	Microsoft Windows XP [Version 5.1.2600] (C) Copyright 1985-2001 Microsoft Corp. C:\Documents and Settings\user>ipconfig
VPC-4> ip dhcp DDORA IP 192.168.2.51/24 GW 192.168.2.1	Windows IP Configuration
VPC-4> show ip	Ethernet adapter Local Area Connection:
NAME : VPC-4[1] IP/MASK : 192.168.2.51/24 GATEWAY : 192.168.2.1 DNS : 8.8.8.8 DHCP SERVER : 172.16.0.1 DHCP LEASE : 93777, 93780/46890/82054 MAC : 00:50:79:66:68:2c LPORT : 20000 RHOST:PORT : 127.0.0.1:30000 MTU : 1500	Connection-specific DNS Suffix . : IP Address. : 192.168.2.52 Subnet Mask : 255.255.255.0 Default Gateway : 192.168.2.1
VPC-4> █	C:\Documents and Settings\user>color f0 C:\Documents and Settings\user> █

Local Area Connection Properties

General | Advanced

Connect using:
Intel(R) PRO/1000 MT Network Contr

Internet Protocol (TCP/IP) Properties

General | Alternate Configuration

Obtain an IP address automatically
Use the following IP address:
IP address:
Subnet mask:
Default gateway:

Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:

Advanced...

OK Cancel

